

Dr. Xiao Cheng

LECTURER (Similar to U.S. Assistant Professor)
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EDUCATION

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|--|-------------------|
| • University of New South Wales (UNSW Sydney) | Sydney, Australia |
| • <i>Ph.D. - Computer Science and Engineering, Software Security Analysis;</i> | 2021 - 2024 |
| • Beijing University of Posts and Telecommunications | Beijing, China |
| • <i>Bachelor & Research Master - Engineering;</i> | 2014 - 2021 |

PUBLICATIONS (CONFERENCE)

(# co-first author, * corresponding author)

- **(ISSTA'2026, CORE-A)**: Jiawei Yang, **Xiao Cheng***, Jiawei Wang, Xiapu Luo and Yulei Sui. Multi-Stage On-Demand Program Slicing for Modular Analysis of Multi-Threaded Programs.
- **(FSE'2026, CORE-A*)**: Linfeng Liang, **Xiao Cheng**, T.Y. Chen and Xi Zheng. From Particles to Perils: SVGD-Based Hazardous Scenario Generation for Autonomous Driving Systems Testing.
- **(S&P'2026, CORE-A*)**: Shangzhi Xu, Ziqi Ding, **Xiao Cheng***, Yuekang Li, Nan Sun, Benjamin Turnbull, Shuangxiang Kan and Siqi Ma. PUFFERDOS: Efficient and Effective Attack String Generation for Regular Expression Denial of Service Vulnerabilities.
- **(FSE'2026, CORE-A*)**: **Xiao Cheng**[#], Zhihao Guo[#], Huan Huo and Yulei Sui. TLR: Codebase-Level C Memory Management Error Repair with Large Language Models.
- **(NDSS'2026, CORE-A*)**: Shuangxiang Kan, **Xiao Cheng***, Yuekang Li. MUTATO: Enhancing Fuzz Drivers with Adaptive API Option Mutation.
- **(ICSE'2026, CORE-A*)**: Shuangxiang Kan, Yuekang Li, **Xiao Cheng***, Yulei Sui. SpecGuru: Hierarchical LLM-Driven API Points-to Specification Generation with Self-Validation.
- **(OOPSLA'2025, CORE-A)**: Jiawei Wang[#], **Xiao Cheng**[#], Yulei Sui. Efficient Abstract Interpretation via Selective Widening.
- **(ECOOP'2025, CORE-A)**: Jiawei Yang[#], **Xiao Cheng**[#], Bor-Yuh Evan Chang, Xiapu Luo and Yulei Sui. Taming and Dissecting Recursions through Interprocedural Weak Topological Ordering.
- **(FSE'25, CORE-A*)**: Haodong Li[#], **Xiao Cheng**[#], Guohan Zhang, Guosheng Xu, Guoai Xu and Haoyu Wang. Mitigating Emergent Malware Label Noise in DNN-Based Android Malware Detection.
- **(ISSTA'2025, CORE-A*)**: Haodong Li, **Xiao Cheng***, Yanjie Zhao, Guosheng Xu, Guoai Xu and Haoyu Wang. Understanding Model Weaknesses: A Path to Strengthening DNN-Based Android Malware Detection.
- **(FSE'2024, CORE-A*, ACM SIGSOFT Distinguished Paper Award)**: **Xiao Cheng**, Jiawei Ren and Yulei Sui. Fast Graph Simplification for Path-Sensitive Typestate Analysis through Tempo-Spatial Multi-Point Slicing.
- **(ICSE'2024, CORE-A*)**: **Xiao Cheng**, Jiawei Wang and Yulei Sui. Precise Sparse Abstract Execution via Cross-Domain Interaction.
- **(ISSTA'2022, CORE-A*)**: **Xiao Cheng**, Guanqin Zhang, Haoyu Wang, and Yulei Sui. Path-Sensitive Code Embedding via Contrastive Learning for Software Vulnerability Detection.
- **(OOPSLA'2020, CORE-A*, ACM SIGPLAN Distinguished Paper Award)**: Yulei Sui, **Xiao Cheng**, Guanqin Zhang, and Haoyu Wang. Flow2Vec: Value-Flow-Based Precise Code Embedding.
- **(ICECCS'2019, CORE-A)**: **Xiao Cheng**, Haoyu Wang, Jiayi Hua, Miao Zhang, Guoai Xu, Li Yi, Yulei Sui. Static Detection of Control-Flow-Related Vulnerabilities Using Graph Embedding.
- **(ICFEM'2023, CORE-C)**: **Xiao Cheng**. Vulnerability Detection via Typestate-Guided Code Representation Learning.

PUBLICATIONS (JOURNAL)

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- **(TOSEM'2024, CORE-A*)**: Jiawei Ren, Yulei Sui, **Xiao Cheng**, Yuan Feng and Jianjun Zhao. Dynamic Transitive Closure-Based Static Analysis through the Lens of Quantum Search.
 - **(TDSC'2022, CORE-A*)**: **Xiao Cheng**, Xu Nie, Ningke Li, Haoyu Wang, Zheng Zheng and Yulei Sui. How About Bug-Triggering Paths? - Understanding and Characterizing Learning-Based Vulnerability Detectors.
 - **(TOSEM'2021, CORE-A*)**: **Xiao Cheng**, Haoyu Wang, Jiayi Hua, Guoai Xu, and Yulei Sui. DeepWukong: Statically Detecting Software Vulnerabilities Using Deep Graph Neural Network.

TEACHING EXPERIENCE

- **COMP3050 Software Systems Development and Operations** Macquarie University
Convenor, Lecturer 2026 S1
- **COMP8700 IT Professional Practice** Macquarie University
Lecturer 2025 S2 & 2026 S1
- **COMP6131 Software Security Analysis** UNSW
Course Admin 2024
- **41128 Software Analysis Studio** On campus
Teaching foundations of software analysis 2021 & 2022 & 2023 & 2024
- **Software Analysis, SSTC Software Engineering Studio** Remote
Teaching foundations of software analysis 2022 & 2023 & 2024
- **41184 Secure Programming and Penetration Testing** On campus
Teaching foundations of software verification 2024 & 2023
- **41181 Information Security and Management** On campus
Teaching IT security policy development and human security management 2024

SERVICES

- Program Committee of FSE 2027, ICSE 2027, ASE 2026, ISSTA 2026, FSE 2026, ISSRE 2025, PAKDD 2026/2025, ACISP 2026, APSEC 2025 (SEIP), LMPL 2025, FSE 2025 (SRC), ISSRE 2024 (DS)
- Artifact Evaluation Committee of PLDI 2026, ICSE 2025, ISSTA 2024/2023, SAS 2023, FormaliSE 2025/2024/2023
- Web Program Chair of LCTES 2024
- Reviewer of SAS 2025, TOSEM, TSE, TDSC, TACO, ASEJ

HONORS AND AWARDS

- 2025, Amazon Research Award (ARA)
- 2025, The Norman Foo Memorial Best Research Paper Prize
- 2024, ACM SIGSOFT Distinguished Paper Award (FSE)
- 2024, Development and Research Training Grant (DRTG), UNSW
- 2024, ACM SIGSOFT CAPS Travel Grant
- 2023, Google ASPIRE Award
- 2021, International Research Training Program Scholarship (IRTP) Offer
- 2020, ACM SIGPLAN Distinguished Paper Award (OOPSLA)

PROJECTS

- **SVF:** A source code analysis framework that enables interprocedural dependence analysis for LLVM-based languages.
- **Teaching Software Analysis:** An online open courses for learning software analysis via SVF.
- **DeepWukong:** a graph neural network based software vulnerability detector.